ADITYA PARIHAR

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PROFILE

Computer Science student targeting **AI Security** roles, with a focus on **Cybersecurity**, **Backend Development**, and **Database Systems**. Hands-on experience in **Python**, **C/C++**, **PostgreSQL**, and privacy-preserving ML for **network intrusion detection**.

EDUCATION

SRM University AP — Andhra Pradesh, India
Bachelors of Science (Hons.) in Computer Science — *Expected 2028*

Shiksha Bharti Public School — Badshapur, Haryana
Intermediate (11th-12th) — *Completed*

Narayana E-Techno School, Sector 37C — Gurugram, Haryana
Secondary Education (10th) — *Completed*

EXPERIENCE

Cybersecurity Intern — Feb 2025 – Mar 2025
LaunchED Global × IIT Bombay (Avenues) — *Remote*

- Completed a 2-month **Cybersecurity Internship** conducted in collaboration with **SJMSOM, IIT Bombay**.
- Gained hands-on exposure to **cybersecurity operations**, **risk analysis**, and industry-standard security practices.
- Developed foundational understanding of security concepts, threat assessment, and cybersecurity workflows through practical learning activities.

Google Gemini Ambassador — 2025
Google Gemini Program — *Campus Ambassador*

- Selected as a **Google Gemini Ambassador (2025)** for representing AI initiatives and technical community engagement activities at the university level.
- Collaborated with student communities to promote awareness of **Generative AI**, **Google Gemini tools**, and emerging AI technologies through discussions, events, and technical learning activities.
- Worked with Google Gemini ecosystem tools and participated in AI-focused learning, collaboration, and technology awareness programs.

RESEARCH & PROJECTS

Hierarchical Federated Learning for Intrusion Detection | *SAINT, PyTorch* **[Ongoing]** — Edge AI Research Lab

- Designed a three-tier (client → edge cluster → root) federated IDS trained across **UNSW-NB15** and **CIC IoMT 2024** without centralizing raw traffic, over 20 non-IID clients.
- Built a leakage-safe preprocessing pipeline (deduplication, split-before-scale, unified 13-class labels) and a two-machine federated deployment over HTTP.

Low-Latency Voice Activation & Streaming ASR | *ESP32-S3, TFLite Micro, PyTorch* — ISRO SIH26172 | GitHub

- Built an edge–server pipeline: wake-word detection on **ESP32-S3** with a 2 s DMA pre-roll buffer, targeting < 256 KB RAM, feeding a causal streaming ASR exported to TFLite.
- Owned **model training and security of the audio stream**.

EdgeCast: Adaptive AI Blending of Weather Forecasts | *Python, LightGBM, FastAPI* — SIH26081 | GitHub

- Co-built a blender that learns per-grid-point trust weights across ML and physical models; best max-temperature RMSE at all 5 lead days and 9–18% lower wind RMSE than any single source.
- Calibrated district alerts (English/Telugu), raising heatwave detection (POD) from 0.32 to 0.90.

Hostel Management Database System | *PostgreSQL, SQL* | GitHub

- Built a normalized relational database using PostgreSQL with tables, relationships, constraints, and ER modeling for hostel operations.

Full Stack Web Application | *Python, Web Technologies, Database Integration* | GitHub

- Developed a full-stack application with scalable backend architecture, database integration, and modern web technologies.

Focused To-Do App | *Python* | GitHub

- Created a Python-based task manager featuring structured task organization, focus tracking, and productivity-oriented workflows.

newtxtext,newtxmath

TECHNICAL SKILLS

Languages: Python, C, C++, SQL
AI/ML: PyTorch, TensorFlow/TFLite Micro, LightGBM, scikit-learn
AI Security: Federated Learning, ML-based Intrusion Detection, TinyML/Edge AI
Security Tools: Wireshark, Nmap, Burp Suite, Kali Linux
Foundations: Data Structures & Algorithms, Computer Architecture, Assembly Basics, Linux

ACHIEVEMENTS

- Completed **Udemy Malware Analysis** course covering malware behavior, analysis fundamentals, and cybersecurity concepts.
- Qualified **MongoDB Certification Exam** demonstrating understanding of NoSQL databases and database management fundamentals.
- Completed **Cybersecurity Mentorship Program** by **BTech Wallah** in association with **LaunchED Global**.
- Led a 6-member team on two **Smart India Hackathon** 2026 problem statements (ISRO)

INTERESTS

- Cybersecurity • AI Security • Hardware Components & System Design • Networking • Backend Development